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DECLARATION

We, Vukona Maritz, Tebogo Thaba and Nhlanhla Nkosi declare that the contents of this project represent our own unaided work, and that the project has not previously been submitted for academic examination towards any qualification. Furthermore, it represents our own opinions and not necessarily those of the Vaal University of Technology.



10/10/2025

Signed



Date

11/10/2025

Signed



Date

11/10/2025

Signed

Date

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AI INVENTORY FORECASTING SYSTEM

AI Solution

This project introduces an AI-powered system designed to forecast product demand for retail businesses. By analyzing historical sales data and time series models, the python-built system predicts future demand with high accuracy. The system particularly addresses the challenges faced by South African SMEs and township retailers, supporting the Fourth Industrial Revolution's goal of democratizing AI technology for inclusive economic growth. This will enable smarter decision-making and cost-effective growth. The system is built using Python.

Problem Definition

Retail businesses in South Africa, especially small and medium enterprises (SMEs), struggle with inaccurate demand forecasting. These inefficiencies are amplified during peak seasons and promotional periods when demand fluctuates unpredictably. Traditional forecasting methods rely on static spreadsheets and intuition, which fail to capture dynamic market behaviour. The lack of intelligent forecasting tools limits scalability and profitability, particularly for township retailers and local supply chains.

Business Objectives

The AI Demand Forecasting System is designed to help South African retailers, especially SMEs and township stores, make smarter business decisions using advanced data analysis and machine learning by predicting product demand more accurately.

Key objectives include:

1. Demand Forecasting Accuracy:

- **Objective:** Achieve 90%+ demand forecasting accuracy within a specified timeframe.
- **Success Criteria:** Regularly validate forecasting accuracy against actual sales data and achieve the target accuracy rate.
- **Relevance:** Accurate forecasting directly impacts profitability, customer satisfaction, and cash flow management for retailers.

2. Financial Impact and ROI for SME Retailers:

- **Objective:** Generate over R1 million in collective savings for participating retailers through improved margins and optimized purchasing.
- **Success Criteria:** Achieve a strong ROI within 18 months and increase average profit margins in all participating stores.
- **Relevance:** Demonstrates clear financial value while supporting sustainable business growth for SMEs and township retailers.

3. Market Penetration and Scalability:

- **Objective:** Deploy the system across 100 retail outlets in Gauteng within 24 months, focusing on township stores and SME chains.
- **Success Criteria:** Achieve 90% user adoption rate and maintain 85% customer satisfaction scores among participating retailers.
- **Relevance:** Ensures broad impact and validates the solution's effectiveness across diverse retail environments.

4. Technology Adoption and Digital Transformation:

- **Objective:** Establish the system as the leading AI demand forecasting platform for South African SME retailers.
- **Success Criteria:** Process over 500 transactions monthly and maintain 99.5% system uptime with sub-3-second response times.

- **Relevance:** Supports 4IR goals by democratizing AI technology access for small businesses and promoting digital transformation.

Requirements

- Develop hybrid forecasting models combining LSTM neural networks, XGBoost, LightGBM and ARIMA for strong demand prediction.
- Accept and process historical sales data (CSV, Excel, or DB).
- Should handle multiple stores and product categories.
- Secure data handling.
- Historical sales data.

Constraints

- The system must comply with South African POPIA; sensitive business data must be encrypted and securely stored.

Risks

- Data Quality Risk: Inconsistent or incomplete sales data could compromise forecast accuracy.
- Adoption Risk: Resistance to change from traditional retailers comfortable with manual processes.
- Technical Risk: Integration challenges with diverse legacy POS systems and accounting software.
- Market Risk: Economic volatility affecting consumer spending patterns and demand forecasting models.

- Scalability Risk: System performance degradation as user base and data volume grow.

Tools and Techniques

- **Machine Learning & Time Series Models:**
 - **ARIMA** for linear, seasonal patterns.
 - **XGBoost** for capturing complex, non-linear relationships.
 - **LightGBM** for fast, high-performance gradient boosting, suitable for large datasets and feature-rich inputs.
- **Deep Learning:**
 - **LSTM** neural networks for sequential 4labelled4.
- Backend: Python (terminal-based)
- ML Libraries: scikit-learn, pandas, NumPy, TensorFlow/Keras
- Database: SQLite or PostgreSQL (accessed via Python libraries)
- Version Control: GitHub with Kanban board and issue tracking
- Natural Language Processing (NLP): Analyze customer reviews for demand signals (using Python NLP libraries)

AI Solution – THEORETICAL ASPECT

Machine Learning Approach

The AI Demand Forecasting System employs a sophisticated multi-model ensemble approach specifically designed to handle the complex, dynamic nature of retail demand patterns in the South African market. Our strategy combines multiple advanced algorithms to create a strong forecasting system capable of adapting to various retail environments and consumer behaviours.

1. Ensemble Forecasting Architecture

- **Objective:** Combine multiple specialized forecasting models to achieve superior accuracy and resilience against model failure.
- **Approach:** We implement a three-tier ensemble system:
 - Tier 1: Time series model – **ARIMA** (Autoregressive Integrated Moving Average):
 - **Rationale for Selection:** ARIMA models are selected due to their established effectiveness in time series forecasting, particularly in capturing linear relationships. Key applications include modelling trends and seasonality within the data.
 - **Functional Description:** ARIMA models analyze time series data based on its historical values. ARIMA models forecast future values by analyzing patterns in past data. It's like looking at a product's sales history to predict how much will sell next month.

The model uses three key elements:

Past Sales (AR): It considers how sales in the past affect current sales. For example, if sales were high last week, they might also be high this week.

Removing Trends (I): It adjusts the data to remove trends or patterns that could skew the forecast. Think of it as labelling the playing field so you can see the true underlying patterns.

Accounting for Errors (MA): It looks at past forecasting errors to improve future predictions. If the model consistently underestimated sales in the past, it will adjust its future forecasts accordingly.

- **Relevance to the Project:** Given the presence of potential trends and seasonal buying behaviours that influence sales, ARIMA is used for analysing and forecasting the time-series data for accurate results.
- Tier 2: Machine learning model – **XGBoost** (Extreme Gradient Boosting):
 - **Rationale for Selection:** XGBoost is selected due to its proven ability to model complex, non-linear feature interactions effectively. As a gradient boosting algorithm, it assembles multiple decision trees into a strong predictive model.
 - **Functional Description:** XGBoost implements gradient boosting, constructing multiple decision trees to build a strong predictive model. Regularization techniques are incorporated to reduce overfitting. The algorithm natively handles missing data.
 - **Relevance to the Project:** Data pertaining to customers and promotions can be incorporated into the XGBoost model, this is expected to increase accuracy.

- Tier 3: Deep learning – **LSTM** (Long Short-Term Memory):
 - **Rationale for Selection:** LSTM networks are chosen because they are specifically designed to capture long-term dependencies in sequential data, particularly where patterns develop across extended time frames.
 - **Functional Description:** LSTM networks, a type of recurrent neural network (RNN), possess memory cells that store data over significant time periods. Input, output, and forget gates govern the flow of data into and out of memory cells.
 - **Relevance to the Project:** Data points spanning extended temporal ranges may be labelled to extrapolate future sales. Seasonal trends are intended to be captured. The impact of past events is intended to be identified.

- Tier 4: Machine learning model: **LightGBM Models**

LightGBM (Light Gradient Boosting Machine) is a highly efficient boosting framework developed by Microsoft that excels in fast, scalable, and high-performance gradient boosting. Its ability to handle large-scale data, extensive feature sets, and categorical variables natively makes it an ideal choice for demand forecasting in complex retail environments.

Key Features & Advantages:

- **Speed & Scalability:**

LightGBM employs histogram-based algorithms and leaf-wise tree growth, enabling it to train faster and handle millions of samples with minimal hardware resources.

- **High Performance with Large Feature Sets:**
It efficiently manages thousands of features, including lag variables, rolling features, categorical variables, and external data, without significant slowdowns.
- **Handling of Missing Data & Categorical Variables:**
Native support for missing values and categorical features reduces preprocessing overhead, leading to cleaner pipelines.
- **Customizable & Regularized:**
Supports early stopping, multiple objective functions, and regularization parameters, aiding in avoiding overfitting.
- **Compatibility & Easy Integration:**
Can be seamlessly integrated into existing Python-based pipelines, compatible with scikit-learn APIs for model training, evaluation, and hyperparameter tuning.
 - **Implementation:** Dynamic weight allocation based on recent performance metrics, with automatic model selection based on data characteristics and forecast horizon.

2. Transfer Learning for SME Adaptation

- **Objective:** Enable rapid model deployment for new retailers with limited historical data.
- **Approach:** Pre-trained models on large retail datasets that can be fine-tuned for specific store characteristics and customer demographics; this allows the system to reuse knowledge from a model trained on a large, rich dataset and apply it to a new, smaller dataset.
- **Implementation:** Employ domain adaptation techniques to transfer knowledge from similar stores while ensuring the preservation of unique store-specific patterns.

Data

The AI Demand Forecasting System utilizes comprehensive, multi-dimensional data sources to ensure accurate predictions across diverse South African retail environments.

- Transaction Records: Point-of-sale data including product SKUs, quantities sold, timestamps, prices.
- Customer Data: Purchase history, loyalty program data, demographic information (anonymized), and return patterns.
- Promotional Data: Marketing campaign schedules, discount structures, price changes, and promotional effectiveness metrics.



Retail Store Inventory Forecasting Dataset

date,store id,product id,category,region,inventory level,units sold,units ordered,demand forecast,price,discount,weather condition,holiday/promotion,competitor pricing,seasonality														
2022/01/01,S001,P0001,Groceries,North,231,127,55,135.47,33.5,20,Rainy,0,29.69,Autumn														
2022/01/01,S001,P0002,Toys,South,204,150,66,144.04,63.01,20,Sunny,0,66.16,Autumn														
2022/01/01,S001,P0003,Toys,West,102,65,51,74.02,27.99,10,Sunny,1,31.32,Summer														
2022/01/01,S001,P0004,Toys,North,469,61,164,62.18,32.72,10,Cloudy,1,34.74,Autumn														
2022/01/01,S001,P0005,Electronics,East,166,14,135,9.26,73.64,0,Sunny,0,68.95,Summer														
2022/01/01,S001,P0006,Groceries,South,138,128,102,139.82,76.83,10,Sunny,1,79.35,Winter														
2022/01/01,S001,P0007,Furniture,East,359,97,167,108.92,34.16,10,Rainy,1,36.55,Winter														
2022/01/01,S001,P0008,Clothing,North,380,312,54,329.73,97.99,5,Cloudy,0,100.09,Spring														
2022/01/01,S001,P0009,Electronics,West,183,175,135,174.15,20.74,10,Cloudy,0,17.66,Autumn														

Model

The AI Demand Forecasting System employs a Hybrid Ensemble Architecture that evaluates multiple forecasting models including ARIMA, LightGBM, XGBoost, and LSTM across each product category and store. For every SKU, the system selects the model with the lowest Mean Absolute Error (MAE), ensuring best accuracy across varying time horizons, seasonal patterns, and market conditions. This fallback logic enables good performance even in unstable retail environments, minimizing forecast error and supporting inventory and pricing decisions.

Model Evaluation for Accuracy

1. Forecasting Accuracy Metrics

- **Purpose:** Measure prediction accuracy across different time horizons and product categories.
- **Method:**
 - MAPE (Mean Absolute Percentage Error): Target <5% for fast-moving products, <10% for slow-moving items.
 - WMAPE (Weighted MAPE): Account for sales volume differences across products.
 - Forecast Bias: Monitor systematic over/under-forecasting tendencies.

2. Business Impact Assessment

- **Purpose:** Measure financial and operational improvements from AI implementation.
- **Method:**
 - Gross Margin Improvement: Track margin increases from optimized purchasing and reduced waste.

- Customer Satisfaction Scores: Monitor product availability and customer experience metrics.
- Cash Flow Optimization: Measure working capital improvements through better inventory management.

3. Model Testing

- **Purpose:** Ensure model performance under various market conditions and data quality scenarios.
- **Method:**
 - Stress Testing: Evaluate performance during holiday seasons, economic disruptions, and supply chain issues.
 - Cross-validation: Time-series cross-validation with expanding windows to test temporal stability.
 - Sensitivity Analysis: Assess model behavior with missing data and outlier scenarios.

Time Series Analysis on Data

The demand forecasting system incorporates sophisticated time series analysis techniques to capture complex temporal patterns in retail sales data, enabling accurate predictions across multiple time horizons and seasonal cycles.

1. Seasonal Decomposition and Pattern Recognition

- Analysis: Multi-level seasonal decomposition identifying daily, weekly, monthly, and annual patterns using STL (Seasonal and Trend decomposition using Loess) and X-13ARIMA-SEATS methods.
- Implementation: Automated detection of seasonal patterns including South African holidays (Heritage Day, Youth Day), payday cycles

(month-end spikes), and cultural events (Christmas, Ramadan) affecting different product categories.

2. Trend Analysis and Regime Change Detection

- Analysis: Long-term trend identification using Hodrick-Prescott filters and structural break detection algorithms to identify significant market shifts.
- Implementation: Real-time monitoring for trend changes caused by economic events, competitor actions, or consumer behavior shifts, with automatic model retraining triggers.

3. Volatility Modelling and Demand Uncertainty

- Analysis: GARCH (Generalized Autoregressive Conditional Heteroskedasticity) models to quantify demand uncertainty and forecast confidence intervals.
- Implementation: Dynamic safety stock calculations based on demand volatility, lead time variability, and service level targets.

4. Multi-variate Time Series Integration

- Analysis: Vector Autoregression (VAR) models capturing interdependencies between related products, cross-selling effects, and cannibalization patterns.
- Implementation: Category-level demand modelling accounting for substitute products, complementary items, and promotional spillover effects.

Solution Techniques

1. Hybrid Ensemble Forecasting: Combines LSTM neural networks, XGBoost gradient boosting, and ARIMA time series models using dynamic weighting based on recent performance and data characteristics.

2. **Transfer Learning for Rapid Deployment:** Pre-trained models on large retail datasets enable quick adaptation to new stores with limited historical data through domain adaptation and few-shot learning techniques.
3. **Automated Feature Engineering:** Machine learning pipelines automatically extract relevant features from raw data including lag variables, rolling statistics, seasonal indicators, and external factor correlations.
4. **Real-time Model Monitoring and Adaptation:** Continuous performance tracking with automated retraining triggers ensures models stay current with changing market dynamics and maintain forecasting accuracy.

Deep Learning

1. **Long Short-Term Memory (LSTM) Networks for Demand Forecasting**
 - **How it Learns:** Processes sequences of historical sales data to identify complex temporal dependencies and patterns that traditional time series methods miss. The LSTM architecture maintains memory of both short-term fluctuations and long-term trends.
 - **Training Process:** Trains on multi-dimensional time series including sales, seasonal trends, economic indicators, and promotional data using supervised learning with sequence-to-sequence prediction. Backpropagation through time optimizes the network to predict future demand based on historical patterns.
2. **Convolutional Neural Networks (CNN) for Pattern Recognition**
 - **How it Learns:** Analyzes sales data as 2D representations (time vs. product features) to detect spatial patterns in demand across product categories, store locations, and customer segments.
 - **Training Process:** Uses labelled sales pattern data to learn visual representations of demand spikes, seasonal trends, and cross-category correlations, enabling detection of similar patterns in new scenarios.

3. Autoencoder Networks for Anomaly Detection

- How it Learns: Learns to reconstruct normal demand patterns and identifies anomalies by detecting high reconstruction errors when processing unusual sales data.
- Training Process: Trains exclusively on normal demand data to learn typical patterns. During operation, s
- Significant deviations from expected reconstructions indicate potential data quality issues, external disruptions, or emerging trends requiring attention.

4. Transformer Networks for Multi-modal Data Integration

- How it Learns: Uses attention mechanisms to integrate diverse data sources (sales, seasons, social media, economic indicators) and identify which factors are most relevant for demand prediction at any given time.
- Training Process: Self-supervised learning on large datasets enables the model to understand relationships between different data modalities and their impact on retail demand patterns.

Other Features: Chatbot

The Retail Intelligence Assistant chatbot is a core component that democratizes AI-driven demand forecasting, giving SME retailers and township store owners an intuitive way to access complex forecasting capabilities without technical expertise.

1. Relevance to the Solution

The chatbot is the primary user interface to the AI system, making advanced demand forecasting accessible to users with varying technical backgrounds:

- **Intelligent Forecasting Insights:** Responds in natural language to questions about demand forecasts and SKU performance.

- **Performance Analytics Communication:** Converts forecasting metrics and business intelligence into clear, actionable recommendations.
- **Training and Guidance:** Provides contextual help and practical best-practice advice for business optimization.

2. Well-Planned and Appropriately Set Up

Advanced NLP and Contextual Understanding:

- **Intent Recognition:** Accurately interprets retail-specific queries including sales performance, supplier issues, and key financial indicators.
- **Context Awareness:** Maintains conversation context and follows up on previous questions to support natural, multi-turn dialogue about complex scenarios.

Intelligent Business Intelligence Integration:

- **Real-time Data Access:** Connects directly to forecasting models and databases to deliver up-to-the-minute insights and recommendations.
- **Predictive Alerts:** Proactively identifies likely issues before they occur.
- **Personalized Recommendations:** Tailors advice to store size, location, product mix, local demand patterns, and historical performance.

3. User-Centric Design for SME Retailers:

- **Simple Query Interface:** Accepts plain-language questions like “What should I order for next week?” or “Why are my sales down this month?”
- **Visual Response Enhancement:** Augments text with charts, trend lines, and visual indicators for faster comprehension.
- **Mobile-Optimized Experience:** Fully functional on smartphones and tablets to support retailers who primarily use mobile devices.

4. **Learning and Adaptation Capabilities:**

- **User Behavior Learning:** Adapts to individual retailer preferences, terminology, and decision patterns over time.
- **Feedback Integration:** Incorporates user feedback to refine accuracy and relevance of responses.
- **Knowledge Base Expansion:** Continuously learns from successful interactions and new data to better handle similar future queries.

5. **Accessibility and Inclusivity Features:**

- **Voice Interaction:** Enables speech-to-text and text-to-speech for hands-free use and accessibility.
- **Simplified Interface:** Clean, intuitive design tailored for users with limited digital literacy.

6. **Business Impact and Measurable Value**

The Retail Intelligence Assistant directly supports business objectives by:

- **Reducing Decision Time:** Store managers obtain instant insights without navigating complex dashboards.
- **Improving Accuracy:** Data-driven recommendations reduce human error in ordering and stocking.
- **Enhancing Adoption:** An intuitive conversational interface increases engagement and system utilization.
- **Supporting Training:** Continuous contextual guidance helps users build better inventory management skills.

AI Solution – PRACTICAL ASPECT

Practical Implementation Overview:

The AI Demand Forecasting System is implemented as a comprehensive solution designed specifically for the diverse needs of South African retail environments. The practical solution demonstrates theoretical concepts through a working system that can handle real-world retail challenges while remaining accessible to SME and township retailers. All components are implemented in Python and designed to be run from the terminal.

Technology Architecture

- Backend: Python-based system using libraries like pandas, scikit-learn, statsmodels, tensorflow, etc.
- Machine Learning Pipeline: Focus on how the Python code manages the model lifecycle (training, validation, deployment).
- Database System: PostgreSQL or SQLite, accessed and managed via Python libraries.

Key Features Demonstrated

- Real-time Demand Forecasting: Processes sales data and external factors to generate hourly, daily, and weekly demand predictions, all implemented in Python.

Pilot Implementation Results

- Focus on metrics and results achieved by running the Python-based system.
- Report on achieved demand forecasting accuracy.

This system serves as a foundation for broader deployment across South Africa's diverse retail landscape.

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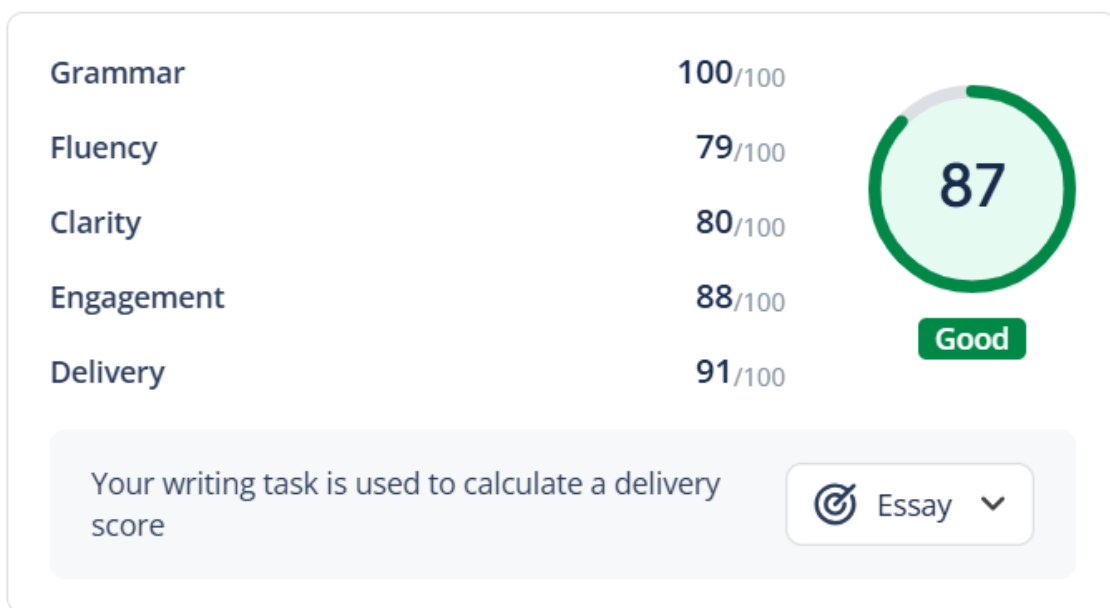
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Grammar Report

This document was proofread using Quillbot's Grammar Checker. Below is the performance summary showing grammar, fluency, Clarity, engagement, and Delivery scores.

Statistics ✕

Writing score ⓘ



Word count

Characters	21550
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Paragraph	195
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